

Vaibhav Katariya

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Education

Jaypee Institute of Technology

B.Tech. in Electronics and Communication Engineering

Noida, India

2023 – 2027

Air Force Bal Bharati School

Senior Secondary Education, Science Stream

New Delhi, India

2010 – 2023

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, PHP

Core CS: Data Structures & Algorithms, Object-Oriented Programming, System Design Fundamentals

Backend: Node.js, Express, FastAPI, REST APIs, JWT, RBAC, Authentication

Databases: PostgreSQL, MongoDB, Firestore, SQL, Query Optimization

Tools & Practices: Git, GitHub, Linux, Nginx, PM2, Idempotent/Retry Design, Debugging

Projects

Autamail | Node.js, Express, Firestore, REST APIs

- Designed backend services and REST APIs for a production email automation platform, processing **1000+ emails** with zero duplication via SHA-256-based deduplication.
- Diagnosed delivery failures under load and fixed them with retry queues and idempotent processing, improving reliability at peak usage.
- Identified inefficient database calls and redesigned the data-access layer with server-side APIs, pagination, and caching, cutting **Firestore query cost by 70%**.
- Implemented JWT authentication and RBAC to secure admin-only operations.

DENTOVAULT | Node.js, Express, PostgreSQL, JWT, Nginx

- Architected backend APIs and a PostgreSQL data model for a medical records platform, designing a hybrid storage approach (PostgreSQL + filesystem) to balance query performance against storage cost for **1800+ high-resolution records**.
- Implemented JWT-based authentication and RBAC for controlled access to sensitive data.
- Deployed and maintained the service on Linux using Nginx and PM2 for stable, ongoing operation.

Interview AI | Node.js, Express, MongoDB, JWT

- Built REST APIs for resume parsing and skill-gap analysis, with validation logic to keep computed results consistent.
- Designed the MongoDB schema and backend auth flow (JWT, file uploads) end-to-end as a solo full-stack build.

Achievements

Winner - Ride Hack 3.0

Dec 2025

Sahyatri – AI-Based Public Transit Occupancy System

- Built FastAPI backend services integrating a YOLOv5 detection pipeline with a live dashboard, achieving **99% accuracy** under real-time constraints on edge hardware.

Position of Responsibility

GDG — Google Developer Groups

Aug 2025 – Present

Technical Lead

- Led development of the club's official website (gdg-jiit.com) and organized BitBox 6.0, driving **2,500+ registrations**.